



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
24-07-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-1

Time : 2:30 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

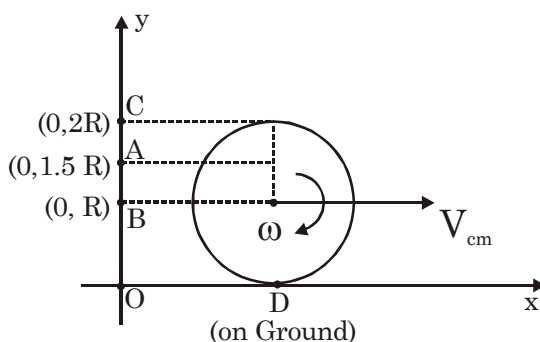
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

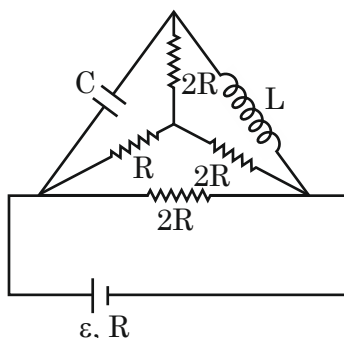
1. A uniform solid disc of radius R , is performing pure rolling on a fixed horizontal surface as shown in figure. O, A, B and C are four points lying on y-axis at the instant shown. Choose the **CORRECT** option(s).



- (A) Angular momentum of disc w.r.t. C and D are different
- (B) Magnitude of angular momentum of disc w.r.t. A (among O, A, B, C and D) is minimum
- (C) Magnitude of angular momentum of disc w.r.t. C (among O, A, B, C and D) is minimum
- (D) Angular momentum w.r.t. O and C are same.

Space for Rough Work

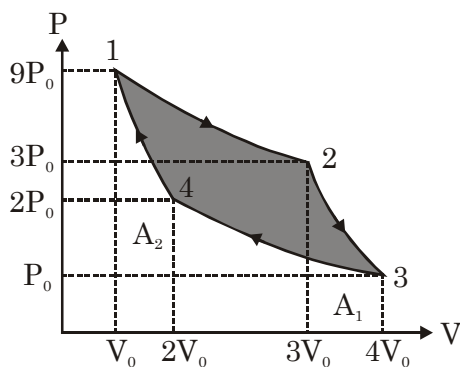
2. A vertical capillary tube with inner radius 0.5 mm, is submerged into water so that the length of its part above the water surface is $h = 25$ mm. Then (surface tension of water = 0.075 N/m, density of water = 10^3 kg/m³, angle of contact = 0°)
- (A) Radius of curvature of meniscus is 0.6 mm.
 (B) Radius of curvature of meniscus is 0.5 mm.
 (C) Height of liquid in the tube is 20 mm.
 (D) Height of liquid in the tube is 25 mm.
3. Figure shows a LCR circuit connected with a dc battery of emf ' ϵ ' and internal resistance ' R '. After a long time (initially the capacitor was uncharged):



- (A) Current through the inductor is $\frac{\epsilon}{8R}$
 (B) Charge stored in the capacitor is $\frac{C\epsilon}{4}$
 (C) Charge stored in the capacitor is $\frac{C\epsilon}{2}$
 (D) Potential difference across the terminals of battery is $\frac{\epsilon}{2}$

Space for Rough Work

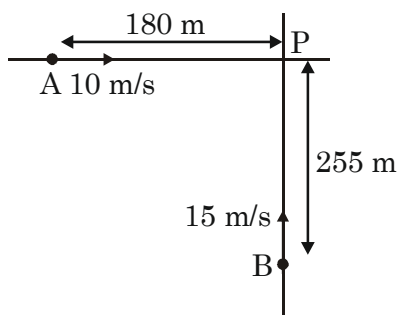
4. A charged particle moves in a certain region of space where there exist mutually perpendicular homogeneous field; electric with intensity E , magnetic with induction B and gravitational field g . It is found that a charged particle is moving with constant velocity in this region. At $t = 0$, electric & magnetic field is switch off & it is found that its velocity begin to decrease just after $t = 0$. Choose the **CORRECT** statement(s)
- (A) After time $\frac{E}{Bg}$, speed of the particle will be minimum
- (B) After $t = 0$, maximum height achieved by the particle will be $\frac{E^2}{B^2g}$
- (C) Keeping net force zero, component of velocity before $t = 0$ in vertical direction will increase on increasing the charge on particle.
- (D) Keeping net force zero, component of velocity before $t = 0$ in horizontal direction will increase on increasing the charge on particle.
5. The figure shows a Carnot cycle for an ideal gas on a $P - V$ diagram. Then (where 1 – 2 and 3 – 4 are isothermal and 2 – 3 and 4 – 1 are adiabatic process) (Note : A_2 and A_1 are the area bounded by curve $4 \rightarrow 1$ & $2 \rightarrow 3$ on V -axis respectively.)



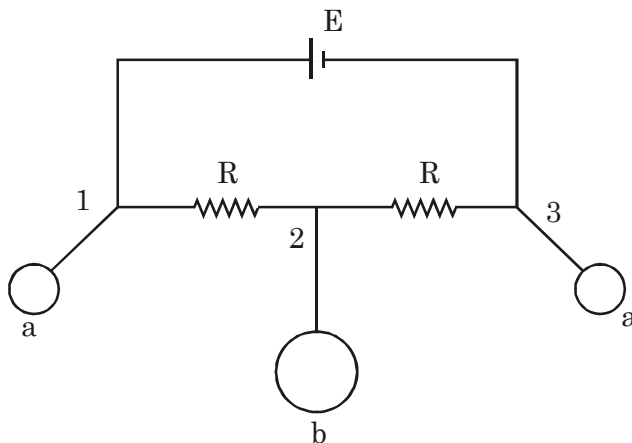
- (A) $\frac{A_1}{A_2} = 1$ (B) $\frac{A_1}{A_2} = 2$ (C) $\frac{A_1}{A_2} = 4$ (D) Data is insufficient

Space for Rough Work

6. Two cars A and B are approaching a crossing P at speeds 10m/s and 15m/s respectively. At the time instant when A is 180 m and B is 255 m from P, A blows a horn of frequency 2.94 kHz. (use speed of sound as 300m/s). Choose the **CORRECT** options :



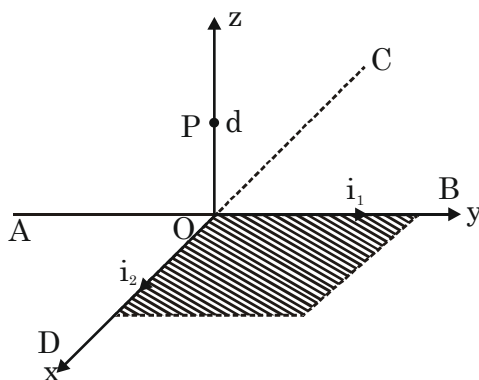
- (A) B hears horn 1 sec after A blows horn. (B) B hears horn instantly after A blows horn.
 (C) Frequency observed by B is 3.12 kHz (D) Frequency observed by B is 3 kHz
7. In the circuit shown in the figure, initially uncharged metal balls with radii a, b & a are connected with points 1, 2 & 3 respectively with long thin conductors. Consider the distance between the balls to be much larger than their size, the charge on the electrical circuit itself and on the connecting conductors is negligible and the internal resistance of battery is zero. Choose the **CORRECT** statement(s). E is emf of the battery.



- (A) Potentials of balls connected at point 1 & 3 are equal.
 (B) Charge on ball of radius b is independent of value of E.
 (C) Charge on each ball is zero
 (D) Charge on ball connected at point 1 is $(4\pi\epsilon_0)aE$.

Space for Rough Work

8. Two identical long conducting wires AOB and COD are placed at right angle to each other, such that 'O' is their common point for the two. The wires carry I_1 and I_2 currents, respectively. Point P is lying at distance 'd' from 'O' along a direction perpendicular to the plane containing the wires. The magnetic field at the point 'P' will be



- (A) in X - Y plane (B) in X - Z plane (C) $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)$ (D) $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)^{1/2}$

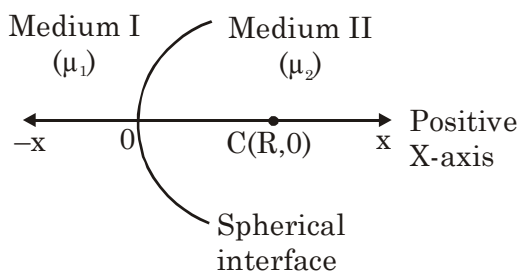
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Four or Five or Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A spherical interface whose radius is $R = 20$ cm separates the two media, I and II as shown in the figure. The refractive index of medium I is μ_1 and that of medium II is μ_2 . Now match the list-I with list-II [Consider the pole 'O' as the origin and principal axis as the X-axis]

**List-I**

- (I) $\mu_2 = 2, \mu_1 = 1$, object distance $u = -10$ cm
 (II) $\mu_2 = 2, \mu_1 = 1$, object distance $u = -20$ cm
 (III) $\mu_2 = 1, \mu_1 = 2$, object distance $u = -40$ cm
 (IV) $\mu_2 = 1, \mu_1 = 2$, object distance $u = -60$ cm

List-II

- (P) $v = -40$ cm
 (Q) $v = -10$ cm
 (R) $v = 40$ cm
 (S) $v = -60$ cm
 (T) $v = \infty$
 (U) $v = -12$ cm

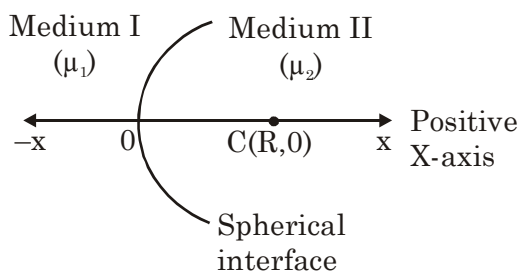
Which of the following option contains all **CORRECT** combinations ?

- (A) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ U
 (B) I $\rightarrow$ P, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ U
 (C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ U
 (D) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ S

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Match the entries on left side with entries on right side. (use $R = 20$ cm)



List-I

- (I) $\mu_2 = 2, \mu_1 = 1$, object distance $u = -10$ cm
- (II) $\mu_2 = 2, \mu_1 = 1$, object distance $u = -20$ cm
- (III) $\mu_2 = 1, \mu_1 = 2$, object distance $u = -40$ cm
- (IV) $\mu_2 = 1, \mu_1 = 2$, object distance $u = -60$ cm

List-II

- (P) $m = 2$
- (Q) $m = -1$
- (R) $m = \infty$
- (S) $m = 0.4$
- (T) $m = 0.5$
- (U) $m = 1$

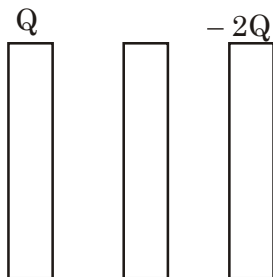
Which of the following option contains all **CORRECT** combinations ?

- (A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ U
- (B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S
- (C) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ S
- (D) I $\rightarrow$ P, II $\rightarrow$ T, III $\rightarrow$ R, IV $\rightarrow$ S

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Match the entries on left side with entries on right side. Three identical metal plates with large surface areas are kept parallel to each other as shown in figure. The left most is given a charge Q , the right most a charge $-2Q$ and the middle one remains neutral. Then



List-I

- (I) The charge appearing on outer surface of right most plate is
(II) The charge appearing on outer surface of left most plate is
(III) The charge appearing on left surface of middle plate is
(IV) The charge appearing on right surface of middle plate is

List-II

- (P) $+\frac{Q}{2}$
(Q) $-\frac{Q}{2}$
(R) $-\frac{3Q}{2}$
(S) $\frac{3Q}{2}$
(T) $+Q$
(U) $-Q$

(A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ U

(B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

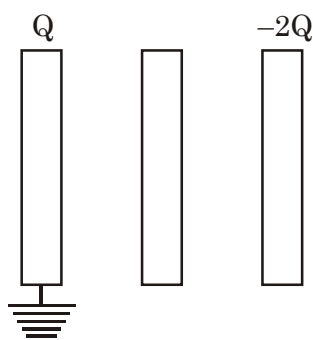
(C) I $\rightarrow$ P, II $\rightarrow$ T, III $\rightarrow$ R, IV $\rightarrow$ U

(D) I $\rightarrow$ Q, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Match the entries on left side with entries on right side when left most plate is grounded. Then



List-I

- (I) The charge appearing on outer surface of right most plate is
 (II) The charge appearing on outer surface of left most plate is
 (III) The charge appearing on left surface of middle plate is
 (IV) The charge appearing on right surface of middle plate is

List-II

- (P) 0
 (Q) Q
 (R) $-Q$
 (S) $2Q$
 (T) $-2Q$
 (U) $3Q$

(A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ U

(B) I $\rightarrow$ P, II $\rightarrow$ P, III $\rightarrow$ T, IV $\rightarrow$ U

(C) I $\rightarrow$ P, II $\rightarrow$ P, III $\rightarrow$ T, IV $\rightarrow$ S

(D) I $\rightarrow$ Q, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

Space for Rough Work

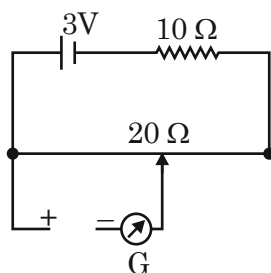
SECTION-II : Numerical Value Type (Up to second decimal place)

No question will be asked in section II

SECTION-III : (Maximum Marks : 18)

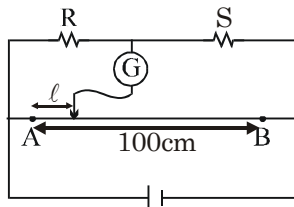
- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A radioactive substance is being produced at a constant rate 'a' per second. Its decay constant is b. The relation between the maximum number of nuclei of X is $N_{\max} = \frac{aK}{b}$, then the value of K is
2. A 10 m long potentiometer wire has a resistance of 20 ohm. It is connected in series with a battery of emf 3V and a resistance of 10 Ω . The internal resistance of cell is negligible. If the length can be read accurately up to 1 mm, the potentiometer can read voltage with an accuracy of $\frac{k}{10}$ (in milli volt). Then the value of k is

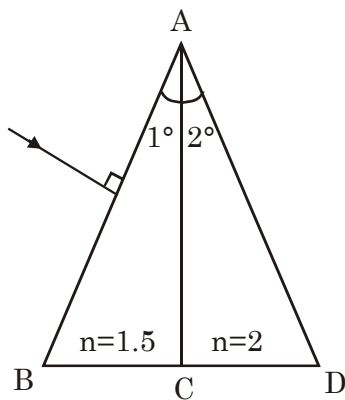


Space for Rough Work

3. Consider the meter bridge circuit without neglecting end corrections. If we used 100Ω and 200Ω resistance in place of R and S respectively, we get null deflection at $\ell_1 = 33.0$ cm. If we interchange the resistances, the null deflection was found to be at $\ell_2 = 67.0$ cm. If α and β are the end correction, then the value of $\alpha + \beta$ should be

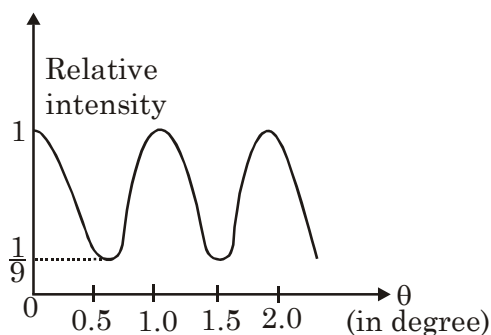


4. A ray of light is incident on a prism as shown in the figure. The total deviation suffered by the ray is $\frac{x^\circ}{2}$. Then find x. ($\angle BAC = 1^\circ$ and $\angle CAD = 2^\circ$, refractive indices are shown in figure.)

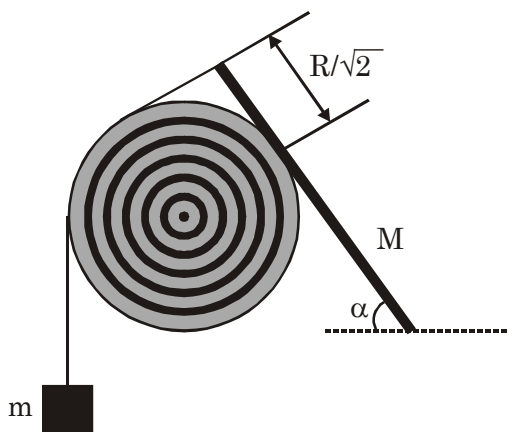


Space for Rough Work

5. In young's double slit experiment relative intensity at a point on the screen may be defined as ratio of intensity at that point to the maximum intensity on the screen. Light of wavelength $7500 \text{ \AA}$ passing through a double slit, produces interference pattern of relative intensity variation as shown in figure. θ on horizontal axis represents the angular position of a point on the screen, then the ratio of amplitudes of the two waves producing interference pattern on the screen is :



6. Through a smooth fixed cylinder of radius R , the axis of which is horizontal, a weightless rope is thrown, to the ends of which a load and a thin uniform rod is attached as shown in the figure. In the position of equilibrium, rod makes an angle $\alpha = 30^\circ$ with the horizontal. The distance from the end of rod to which the rope is attached, to the point of contact of the rod and the cylinder is $\frac{R}{\sqrt{2}}$. If mass M of rod is 6 kg , then find the value of mass m in kg .



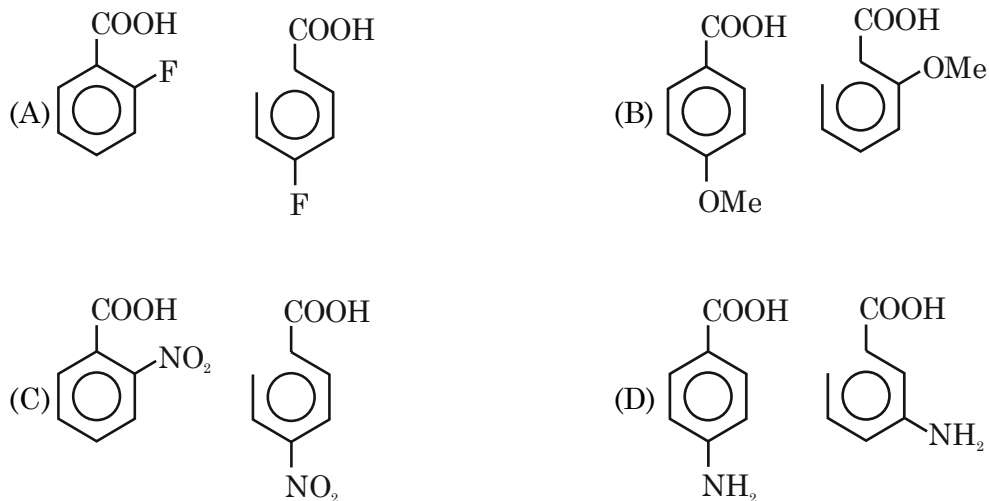
Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 32)**

- This section contains **EIGHT** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.
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1. An ideal gaseous sample at initial state i (P_0, V_0, T_0) is allowed to expand to volume $2V_0$ using two different process; in the first process the equation of process is $PV^2 = K_1$ and in second process the equation of the process is $PV = K_2$. Then,
- (A) Work done in first process will be greater than work in second process (magnitude wise)
- (B) The order of values of work done can not be compared unless we know the value of K_1 and K_2 .
- (C) Value of work done (magnitude) in second process is greater in above expansion irrespective of the value of K_1 and K_2
- (D) 1st process is not possible

Space for Rough Work

2. In which of the following pairs IInd compound is more acidic than the Ist one

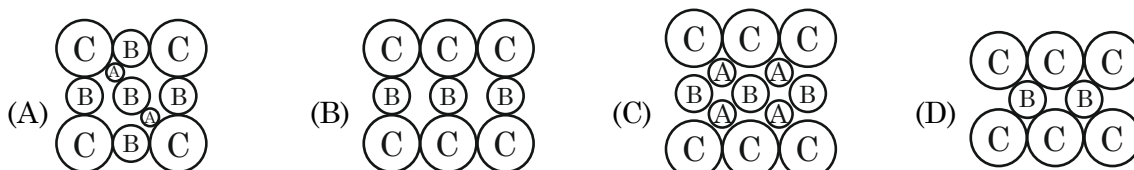
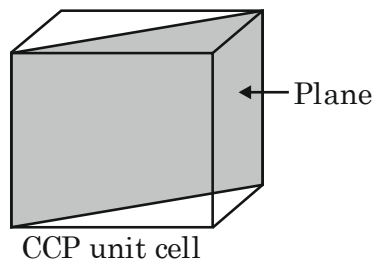


3. Select **CORRECT** statement(s)

- (A) Stability of peroxides and superoxides of alkali metals increases with increase in size of the metal ion
- (B) Increase in stability of peroxides and superoxides in I(A) is due to stabilisation of large anions by larger cations through lattice energy effects.
- (C) The low solubility of LiF is due to its high lattice energy whereas low solubility of CsI is due to smaller hydration energy
- (D) NaOH crystals are not moisture absorbing in nature.

Space for Rough Work

4. In a hypothetical solid C atoms form CCP lattice with A atoms occupying all the tetrahedral voids and B atoms occupying all the octahedral voids. A and B atoms are of the appropriate size such that there is no distortion in the CCP lattice. Now if a plane is cut (as shown) then the cross section of the plane would not be like.



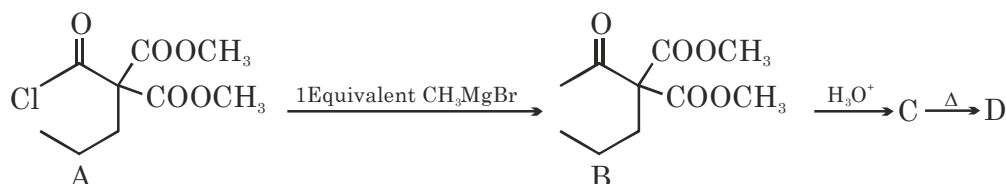
5. Boyle's temperature of a vander Waal's gas is 350 K. Which of the following statement(s) is/are correct for the gas?
- (A) The gas can't be liquefied at 300 K
- (B) The gas can be liquefied at 150 K
- (C) The gas shows negative deviation at low pressure and 275 K
- (D) The gas shows positive deviation at any pressure at 400 K

Space for Rough Work

6. Which of the following specie/molecule has sp^3d^2 hybridised central atom with permanent dipole moment?

- (A) XeO_6^{4-} (B) XeO_2F_2 (C) XeF_5^+ (D) SeF_5^-

7. For the given conversion correct option among the following is



- (A) Maximum number of RMgX molecule that can be consumed by one molecule of A is 6
 (B) Total number of CO_2 molecule release on heating one molecule of C is 2
 (C) One molecule of C can consumed atleast 3 molecule of RMgX .
 (D) Final product D is an aldehyde and can consume only 1 molecule of RMgX

8. Which of the following statement(s) is/are **FALSE**?

- (A) Fe^{3+} gives red precipitate with dimethyl glyoxime in alkaline solution
 (B) Cu^{2+} ion with excess KI gives a brown appearance, which turns into white ppt. on adding hypo solution
 (C) A filter paper soaked in mercurous nitrate turns black in contact with ammonia gas
 (D) Ag^+ gives white precipitate with BaBr_2

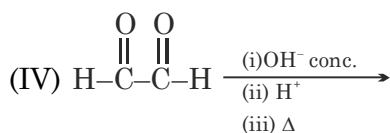
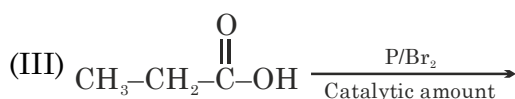
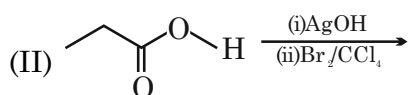
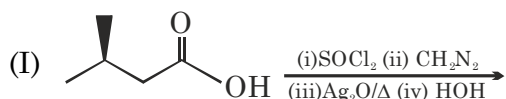
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Four or Five or Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I



List-II

- (P) Substrate is unable to evolve CO_2 on treatment with NaHCO_3 .
- (Q) No dehydration but decarboxylation takes place in the reaction.
- (R) Racemic mixture obtained in the final product
- (S) Parent chain length increases in the final product
- (T) Reactant can undergo decarboxylation on heating.
- (U) Product obtained with lesser number of C atom as compare to reactant.

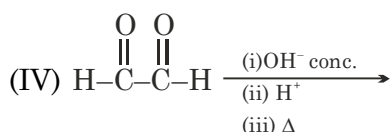
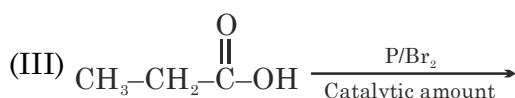
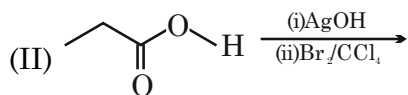
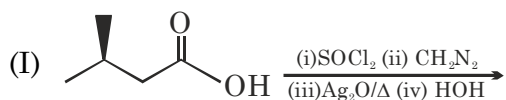
Which of the following option contains all correct combinations ?

- (A) I-P, II-Q, III-T, IV-R (B) I-P, II-Q, III-R, IV-T
 (C) I-S, II-U, III-R, IV-P (D) I-S, II-U, III-P, IV-R

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I



List-II

- (P) Substrate is unable to evolve CO_2 on treatment with NaHCO_3 .
- (Q) No dehydration but decarboxylation takes place in the reaction.
- (R) Racemic mixture obtained in the final product
- (S) Parent chain length increases in the final product
- (T) Reactant can undergo decarboxylation on heating.
- (U) Product obtained with lesser number of C atom as compare to reactant.

Which of the following statement is not true. (about the substrate/reaction given above)?

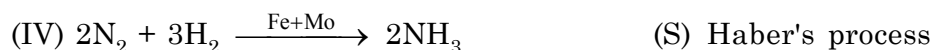
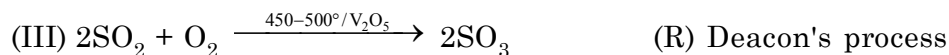
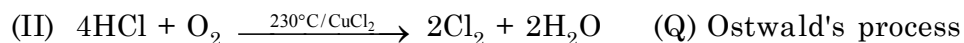
- (A) Arndt eistert reaction is a method of producing β -amino acid from α -amino acid
- (B) Boradine Hunsdiecker reaction involve both dehydration and decarboxylation
- (C) HVZ reaction involve ascent in parent carbon chain
- (D) Glyoxal gives +ve tollen's test but –ve fehling's solution test

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I (Chemical reaction)

List-II (Name of process)



(T) Dow's process

(U) Hall-heroult process

Which of the following option contains all **CORRECT** combinations ?

(A) I-P, II-Q, III-T, IV-R

(B) I-Q, II-R, III-P, IV-S

(C) I-P, II-S, III-R, IV-U

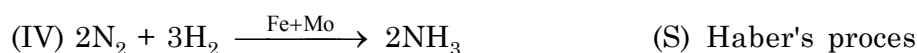
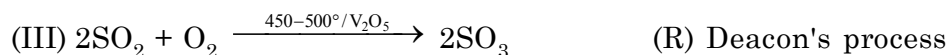
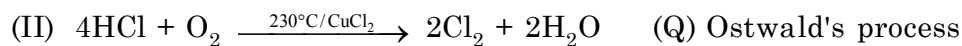
(D) I-P, II-U, III-Q, IV-S

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I (Chemical reaction)

List-II (Name of process)



(T) Dow's process

(U) Hall-heroult process

Which of the following option consists of both **INCORRECT** combination ?

(A) I-Q, II-R

(B) I-Q, IV-S

(C) IV-S, III-P

(D) III-U, IV-T

Space for Rough Work

SECTION-II : Numerical Value Type (Up to second decimal place)

No question will be asked in section II

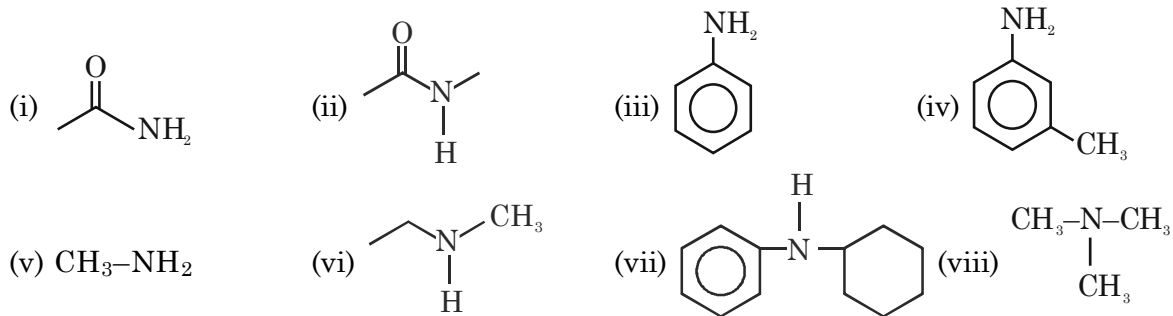
SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. An ideal gaseous mixture of ethane (C_2H_6) and ethene (C_2H_4) occupies 120 litre at 300K, 1atm. The mixture reacts completely with 528 gm O_2 to produce CO_2 and H_2O . Mole fraction at C_2H_6 in the mixture is. (Report answer after multiplying with 10)
($R = 0.08$ litre-atm/K/mole)
2. V_1 Litre of solution A (Resistance = 50 ohm) is mixed with V_2 Litre of solution B (Resistance = 100 ohm). Resistance of final solution is 80 ohm then $\frac{V_2}{V_1}$ will be (using same cell)
3. Correct statement out of the following for penicillin is/are
 - (a) It gives +ve Haloform test
 - (b) It can evolve CO_2 on treatment with $NaHCO_3$
 - (c) It gives +ve 2, 4 DNP test (Braddy's test)
 - (d) Lassaigne extract of penicillin gives -ve test with $AgNO_3$
 - (e) Lassaigne extract of penicillin gives +ve test with $Fe_4[Fe(CN)_6]_3$
 - (f) Lassaigne extract of penicillin gives +ve sodium nitroprusside test
 - (g) Presence of $\begin{array}{c} O \\ || \\ -C-N- \\ | \\ H \end{array}$ amide linkage/gp.
 - (h) It gives -ve test with neutral $FeCl_3$ solution.

Space for Rough Work

4. How many of the following given option will give positive isocyanide test



5. Total number of geometrical isomers for the complex: $[\text{Ir}(\text{acac})(\text{SCN})(\text{H}_2\text{O})(\text{NH}_3)(\text{Py})]^+$

6. For coagulation of 500 mL of $\text{Fe}(\text{OH})_3$ sol in two hours, 1 mL of 1 M Na_2SO_4 electrolyte is required. What is the coagulating value of Na_2SO_4 electrolyte?

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 32)**

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Consider $f(x) = \begin{cases} \sin(\cos x) + \cos(\sin x) & , -\frac{\pi}{2} \leq x \leq 0 \\ \sin(a(1+x)) + 1-x & , 0 < x < \frac{\pi}{2} \end{cases}$, then

- (A) The sum of maximum and minimum values of $f(x)$ in $x \in \left[-\frac{\pi}{2}, 0\right]$ is greater than 2
- (B) The absolute difference of maximum and minimum values of $f(x)$ in $x \in \left[-\frac{\pi}{2}, 0\right]$ is greater than 1
- (C) There is exactly one integer 'a', $a \in \left(0, \frac{\pi}{2}\right)$ for which $f(x)$ has local maximum at $x = 0$
- (D) There are exactly two integers 'a', $a \in (0, \pi)$ for which $f(x)$ has local maximum at $x = 0$

Space for Rough Work

2. If $T(3, 2)$ is the foot of perpendicular drawn from focus $S(2, -1)$ on a tangent to some parabola and its directrix passes through $P(0, 9)$ then
- (A) length of latus-rectum of parabola is $8\sqrt{2}$
- (B) equation of tangent at vertex of parabola is $x + y = 5$
- (C) equation of axis of parabola is $x - y = 3$
- (D) distance of directrix of parabola from its focus is $2\sqrt{2}$
3. If L_1 is the line of intersection of planes $2x - 2y + 3z - 2 = 0$ and $x - y + z + 1 = 0$ and L_2 is line of intersection of planes $x + 2y - z - 3 = 0$ and $3x - y + 2z - 1 = 0$ and P is the plane containing L_1 and L_2 then
- (A) Shortest distance of plane 'P' from origin is $\frac{1}{2\sqrt{2}}$
- (B) Shortest distance of plane 'P' from origin is $\frac{1}{3\sqrt{2}}$
- (C) Volume of tetrahedron formed by plane 'P' with co-ordinate axes is $\frac{9}{784}$ cubic unit
- (D) Plane 'P' is parallel to the line $\frac{x-3}{4} = \frac{4-y}{4} = \frac{z+1}{-7}$

Space for Rough Work

4. Consider $f(x) = \begin{cases} |x|^a \sin\left(\frac{1}{x}\right) + |x|^b (\tan x)^c, & x \neq 0 \\ 0, & x = 0 \end{cases}$, where $a, b, c \in \mathbb{N}$.

If $f(x)$ is differentiable at $x = 0$ then ordered triplet (a, b, c) can be

- (A) (2, 1, 3) (B) (3, 2, 1) (C) (1, 2, 3) (D) (2, 1, 2)

5. Consider $f(x) = \tan x + \tan^2 x \cdot \tan 2x$ and $g(n) = \sum_{r=0}^{n-1} f(2^r)$

Identify the correct statement(s).

- (A) Number of solutions of the equation $f(x) = 1$ in $x \in (-\pi, \pi)$ is 2
 (B) Number of solutions of the equation $f(x) = 1$ in $x \in (-\pi, \pi)$ is 6
 (C) If $g(2020) = \tan(2^m) - \tan(2^r)$ then $(m + r)$ is 2019
 (D) If $g(2020) = \tan(2^m) - \tan(2^r)$ then $(m + r)$ is 2021

6. Let 'B' is a skew symmetric matrix of order 3×3 with real entries. Given $(I + B)$ and $(I - B)$ are non-singular matrices and $A = (I + B)(I - B)^{-1}$ (where $\det(A) > 0$). Identify the correct statement(s).

- (A) $\det(2A) = 8$ (B) $A \cdot A^T = I$ (C) $\det(\text{adj}(A)) = 1$ (D) $A = A^{-1}$

[$\det(P)$ denotes determinant of square matrix P and $\det(\text{adj}(P))$ denotes determinant of adjoint of square matrix P]

Space for Rough Work

7. Given $p, q \in \{0, 1, 2, 3, \dots, 19, 20\}$, consider the system of equations

$$x + y + z = 4$$

$$2x + y + 3z = 6$$

$$x + 2y + pz = q$$

Let

L denotes number of ordered pairs (p, q) so that system has a unique solution

M denotes number of ordered pairs (p, q) so that system has no solution

N denotes number of ordered pairs (p, q) so that system has infinite solutions,
then :

(A) $L = 420$ (B) $M = 20$ (C) $N = 2$ (D) $\frac{L+M+N}{21} = 21$

8. Identify the correct statement(s)

(A) If $f(x) > 0$, $f''(x) > 0 \forall x \in [0, 1]$ and $f(0) = 0$, $f(1) = 1$ then $f(x) \cdot f^{-1}(x) < x^2 \forall x \in (0, 1)$

(B) If $f(x)$ is a twice derivable function and $f(\alpha) = f(\beta) = 0$, $(\alpha < \beta)$, then $(f'(x))^2 + f(x) \cdot f''(x) = 0$ has at least two real roots in $x \in (\alpha, \beta)$

(C) Let $g : [a, b] \rightarrow \mathbb{R}$ be twice differentiable function such that $g''(x) + g'(x) \cdot g(x) = g(x) \forall x \in [a, b]$ and $g(a) = g(b) = 0$ then $g(x) = 0 \forall x \in [a, b]$

(D) Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = 0$ and $f(x) > 0 \forall x \in (-1, 1) - \{0\}$,

then $\frac{f'(c)}{f(c)} = \frac{3}{2} \frac{f'(1-3c)}{f(1-3c)} + \frac{1}{1-2c}$ for atleast one $c \in \left(\frac{1}{3}, \frac{1}{2}\right)$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Four or Five or Six** entries (P), (Q), (R), (S), (T) and (U)
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

In a tournament there are 12 players $S_1, S_2, \dots, S_{12}$. They are divided in 6 pairs at random. From each game a winner is decided on the basis of game played between the two players of the pair. Assume all players are of equal strength.

List-I		List-II
(I) Probability that S_1 is among the losers is	(P)	$\frac{10}{11}$
(II) Probability that exactly one of S_2 and S_5 is among the losers	(Q)	$\frac{1}{2}$
(III) Probability that both S_3 and S_5 are among the winners	(R)	$\frac{5}{22}$
(IV) Probability of S_6 and S_7 not playing against each other	(S)	$\frac{6}{13}$
	(T)	$\frac{6}{11}$
	(U)	$\frac{9}{11}$
Which of the following is INCORRECT combination ?		
(A) (I), (Q)	(B) (II), (T)	(C) (III), (R)
		(D) (IV), (U)

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

In a tournament there are 12 players $S_1, S_2, \dots, S_{12}$. They are divided in 6 pairs at random. From each game a winner is decided on the basis of game played between the two players of the pair. Assume all players are of equal strength.

List-I

List-II

(I) Probability that S_1 is among the losers is

(P) $\frac{10}{11}$

(II) Probability that exactly one of S_2 and S_5 is among the losers

(Q) $\frac{1}{2}$

(III) Probability that both S_3 and S_5 are among the winners

(R) $\frac{5}{22}$

(IV) Probability of S_6 and S_7 not playing against each other

(S) $\frac{6}{13}$

(T) $\frac{6}{11}$

(U) $\frac{9}{11}$

Which of the following is **CORRECT** combination ?

(A) (I), (T)

(B) (II), (S)

(C) (III), (R)

(D) (IV), (Q)

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Let $x + y = 2$, and $x - y = 0$ be the tangent at vertex and axis of a parabola respectively. Straight line $2x - y - 4 = 0$ is a tangent to this parabola.

List-I

List-II

(I) If focus of parabola is (a, b) then $\frac{3(a+b)+5}{3}$ is (P) 6

(II) If length of latus rectum of parabola is $\frac{a\sqrt{2}}{b}$, where (Q) 0

a and b are co-prime numbers, then $|a - b|$ is

(III) If equation of directrix of parabola is (R) 4

$ax + by - 8 = 0$ then $|a + b|$ is

(IV) If image of focus in tangent line $2x - y - 4 = 0$ is (a, b) (S) 1

then $|a - b + 1|$ is

(T) 3

(U) 5

Which of the following is **INCORRECT** combination ?

(A) (I), (T)

(B) (II), (S)

(C) (III), (Q)

(D) (IV), (U)

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

Let $x + y = 2$, and $x - y = 0$ be the tangent at vertex and axis of a parabola respectively. Straight line $2x - y - 4 = 0$ is a tangent to this parabola.

List-I

List-II

(I) If focus of parabola is (a, b) then $\frac{3(a+b)+5}{3}$ is (P) 6

(II) If length of latus rectum of parabola is $\frac{a\sqrt{2}}{b}$, where (Q) 0

a and b are co-prime numbers, then $|a - b|$ is

(III) If equation of directrix of parabola is (R) 4

$ax + by - 8 = 0$ then $|a + b|$ is

(IV) If image of focus in tangent line $2x - y - 4 = 0$ is (a, b) (S) 1

then $|a - b + 1|$ is

(T) 3

(U) 5

Which of the following is **CORRECT** combination ?

(A) (I), (U)

(B) (II), (S)

(C) (III), (R)

(D) (IV), (P)

Space for Rough Work

SECTION-II : Numerical Value Type (Up to second decimal place)

No question will be asked in section II

SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

- 1 If $f(x) = \left(\frac{\pi}{2} - \cos^{-1}(\cos x)\right) \left(\frac{\pi}{2} - \sin^{-1}(\sin x)\right)$, where $x \in [-2\pi, 2\pi]$ then number of integers in the range of $f(x)$ is_____.
2. Let the function $f: \{1, 2, 3, \dots, 2020\} \rightarrow \{2021, 2022, 2024\}$ satisfies $f(1) + f(2) + f(3) + \dots + f(2020) = \text{even integer}$. If number of such possible functions is N , then unit's digit of 'N' is_____.
3. If solution of differential equation $(3\tan x + 4\cot y - 7)\sin^2 y dx - (4\tan x + 7\cot y - 5)\cos^2 x dy = 0$ is $\frac{3}{2}\tan^2 x - \alpha \tan x + \frac{7}{2}\cot^2 y - \beta \cot y + \gamma(\tan x \cdot \cot y) = C$, where α, β, γ are constants and C is arbitrary constant, then $(\alpha + \beta - \gamma)$ is_____.

Space for Rough Work

4. Consider two polynomials $f(x)$ and $g(x)$ given by $f(x) = \sum_{r=0}^{200} \beta_r x^r$ and $g(x) = \sum_{r=0}^{200} \alpha_r x^r$ such that $\beta_r = 1 \forall 100 \leq r \leq 200$ and $f(x+1) = g(x) \forall x \in \mathbb{R}$, then $\sum_{r=100}^{200} \alpha_r$ is equal to 2^m . The value of $\left(\frac{m}{50}\right)$ is ____.
5. If $\int_{-3\pi}^{3\pi} (1 + \cos x + \cos 2x + \dots + \cos 2020x) \cdot (1 + \sin x + \sin 2x + \dots + \sin 2020x) dx = K\pi$, then K is ____.
6. If $\hat{a}, \hat{b}$ are unit vectors and $\vec{c}$ is such that $\vec{c} = \vec{a} \times \vec{c} + \vec{b}$, then maximum possible value of $2[\vec{a} \vec{b} \vec{c}]$ is ____.

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work